

Policy Transition Risk, Carbon Premiums, and Asset Prices

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All files can be run in MATLAB. The files to be run are labelled as Code_MainFile.m and initiate the respective solution and simulation procedure. Notice that the running time can vary significantly depending on the model specification and the chosen grid size (can be adjusted in lines 33 to 44 of the main files) and the number of Monte-Carlo simulations (line 47). In particular, the codes Section 5 have a significantly longer running time than those for Section 4. The zip-file contains two folders, each of them contains various auxiliary functions and procedures that implement all the relevant equations and solution algorithms:

1) Codes_sec4

- Running Code_MainFile.m makes the simulation Figures 1, 2, and D.9 for the benchmark calibration and sample path 135 (can be changed in Code_PlotSelection.m, line 3) as in the paper.
- It also makes the figures for the policy functions D.5, D.6, and D.7.
- It makes the histogram plot D.8 (you need to increase the number of paths to about 200,000 to get smooth histograms, see line 47 in Code_MainFile.m).

Code Variants for Robustness

- Changing transition='off' to transition=on; (see line 100 Code_MainFile.m), runs the model without political transition risk.
 - Y0=1; corresponds to a pure BAU scenario and replicates the left-hand panels of Figure D.3 and Figure D.4 of the online appendix
 - Y0=2; corresponds to a pure PIGOU scenario and replicates Figure E.11 of the online appendix
 - Y0=3; corresponds to a pure CAP scenario and replicates Figure E.12 of the online appendix
- Switching on externalities one-by-one can be done in
 - Disasters: Line 22 (ClimateDisasters='off'; to ClimateDisasters='on';)
 - Tipping Points: Line 97 (Tipping='off'; to Tipping='on';)
- Turning to a tighter temperature cap can be done in line 120 of Code_Initiation.m by changing budget.temperature=2; to budget.temperature=1.5; This replicates Figure E.10 of the online appendix
- Figures without sample paths can be made by using Code_PlotsSelection.m instead of Code_PlotPaths.m

2) Codes_sec5

- Running Code_MainFile.m makes Figures 3 and 4 as well as some additional material.
- In particular, it makes the calibration figures C.1 and C.2 of the online appendix

In case of questions regarding the codes, please contact Christoph Hambel, C.Hambel@tilburguniversity.edu.